

Cybersecurity Domain

Powered By HCL GUVI

Course type	Offline
Total Contact Hours	<i>45 Hours</i>

Course learning objectives

The objective of the course is,

- 1 Understand the fundamental concepts of Data Structures, algorithm analysis, and their applications in Cybersecurity.
- 2 Apply appropriate data structures and algorithmic techniques to solve security-related problems, optimize threat detection, and protect digital systems.
- 3 Apply appropriate data structures and algorithmic techniques to solve security-related problems, optimize threat detection, and protect digital systems.

Unit 1: Basic Data Structures (9 Hours)

- Introduction to Data Structures - Arrays (1D & Multi-dimensional) - String Operations & String Algorithms - Linked Lists (Singly, Doubly & Circular) - Stacks and Queues - Applications of Stacks & Queues - Time Complexity - Space Complexity - Big-O Notation

Hands-on Session:

- Implement Arrays and Linked Lists.
- Stack and Queue using Arrays and Linked Lists.
- Analyze Time & Space Complexity.

Unit 2: Hashing & Trees (9 Hours)

- Hash Tables - Hash Functions - Collision Resolution Techniques - HashMaps - Binary Trees - Binary Search Trees (BST) - Tree Traversals (Inorder, Preorder, Postorder) - Heaps - Priority Queues - Applications of Trees

GUVI

Hands-on Session:

- Implement Hash Tables.
- Binary Search Tree Operations.
- Heap Implementation.
- Tree Traversal Programs.

Unit 3: Graphs & Traversals (9 Hours)

- Graph Representation (Adjacency Matrix & Adjacency List) - Breadth First Search (BFS) - Depth First Search (DFS) - Shortest Path Algorithms (Dijkstra & Bellman-Ford) - Topological Sorting - Minimum Spanning Tree (Prim's & Kruskal's) - Graph Applications

Hands-on Session:

- BFS and DFS Implementation.
- Dijkstra's Algorithm.
- Bellman-Ford Algorithm.
- Minimum Spanning Tree.

Unit 4: Advanced Algorithms (9 Hours)

- Sorting Algorithms (Bubble, Selection, Insertion, Merge, Quick, Heap) - Searching Algorithms (Linear & Binary Search) - Greedy Algorithms - Dynamic Programming - 0/1 Knapsack - Longest Common Subsequence (LCS) - Backtracking - Algorithm Analysis

Hands-on Session:

- Implement Sorting Algorithms.
- Binary Search.
- Dynamic Programming Problems.
- Backtracking Problems (N-Queens, Sudoku).

Unit 5: Cybersecurity Applications and Coding Practice (9 Hours)

GUVI

- Cybersecurity Applications (Pattern Matching, Attack Graph Analysis, Cryptography Algorithms) + Weekly CodeKata Practice

Hands-on Session:

- Perform Data Aggregation using Efficient Data Structures.
- Develop a Simple Recommendation System.
- Analyze Time-Series Data using Appropriate Algorithms.
- Implement Data Processing Pipelines.
- Weekly AI & Data Science CodeKata Challenges.

Course Outcome (COs)

At the end of the course, the student will be able to:

Learning Levels: Re - Remember; Un - Understand; Ap - Apply; An - Analysis; Ev - Evaluate; Cr - Create		Learning Level
1	Understand the fundamental concepts of data structures and their applications in Cybersecurity.	Un
2	Implement efficient data structures to develop secure and optimized cybersecurity solutions.	Ap
3	Analyze graph algorithms, hashing techniques, and tree structures for network security and threat detection.	An
4	Evaluate algorithmic approaches for intrusion detection, secure authentication, and cryptographic applications.	Ev
5	Design secure software solutions by integrating efficient data structures, graph algorithms, and cybersecurity techniques.	Cr